

Regression model for prediction individuals' creditworthiness

Column	Description
CLIENTNUM	Client number. Unique identifier for the customer holding the account
Customer_Age	Demographic variable - Customer's Age in Years
Gender	Demographic variable - M=Male, F=Female
Dependent_count	Demographic variable - Number of dependents
Education_Level	Demographic variable - Educational Qualification of the account holder
Marital_Status	Demographic variable - Married, Single, Divorced, Unknown
Income_Category	Demographic variable - Annual Income Category of the account holder
Card_Category	Product Variable - Type of Card (Blue, Silver, Gold, Platinum)
Months_on_book	Period of relationship with bank
Total_Relationship_count	Total no. of products held by the customer
Months_Inactive_12_mon	No. of months inactive in the last 12 months
Contacts_Count_12_mon	No. of Contacts in the last 12 months
Credit_Limit	Credit Limit on the Credit Card
Total_Revolving_Bal	Total Revolving Balance on the Credit Card
Total_Amt_Chng_Q4_Q1	Change in Transaction Amount (Q4 over Q1)
Total_Trans_Amt	Total Transaction Amount (Last 12 months)
Total_Trans_Ct	Total Transaction Count (Last 12 months)
Total_Ct_Chng_Q4_Q1	Change in Transaction Count (Q4 over Q1)
Avg_Utilization_Ratio	Average Card Utilization Ratio

*Change in Transaction Count (Q4 over Q1): measures how the total number of transactions in the fourth quarter (Q4) of a given year compares to the first quarter (Q1) of the same year. It answers whether transaction volume went up, down, or stayed the same from Q1 to Q4.

1.missing values:

The married status has 25% missing values but we can't erase 25% of our data so we replace missing values with mode.

The missing values in Months_on_book and Total_Relationship_Count are very small so we can erase the rows with missing values.

For income_category, if we change ranges to a number we lose a lot of information, like two people with 62k salary and 78k salary are treated the same way and the difference between two numbers are a lot. So we replace missing values with mode.

To fill missing values in Income_Category, a conditional method based on Credit_Limit binning was used. First, Credit_Limit was divided into four categories (Very Low, Low, High, Very High). Then, for each bin, the mode (most frequent value) of Income_Category was calculated and assigned to rows with missing values in that same bin. This method preserves the relationship between income and credit limit better than using a global mode.

For the rest of the septs we need to split data to train and test .

2.Assessing Normality

1. Shape Metrics: Skewness and Kurtosis

Before running formal statistical tests, we analyzed the shape parameters of each feature's distribution to visually and numerically evaluate its departure from a standard bell curve.

- **Skewness:** This metric measures the symmetry of the data distribution around its statistical mean. A perfectly symmetrical normal distribution has a skewness value of zero. In our dataset, features like the credit limit exhibited high positive skewness, meaning the data has a long tail stretching toward the higher values on the right side.
- **Kurtosis:** This metric measures the "tailedness" of the distribution, which directly indicates the presence and impact of extreme values or outliers. A standard normal distribution has an excess kurtosis of zero. High positive values in our features indicated heavy tails and a sharp central peak, proving that the dataset inherently contains strong anomalies.

2. Formal Statistical Testing

To validate our findings with mathematical certainty, we conducted two rigorous hypothesis tests using Python's `scipy.stats` library. For both tests, we established a strict framework:

- **Null Hypothesis (\$H_0\$):** The underlying data is normally distributed.
- **Alternative Hypothesis (\$H_1\$):** The data does not follow a normal distribution.

If the test calculates a **p-value less than our significance threshold of 0.05**, we reject the null hypothesis, mathematically proving that the data is non-normal.

- **The Shapiro-Wilk Test:** This test calculates a statistic by comparing the ordered sample observations against generalized weights derived from a standard normal distribution. It is highly sensitive and ideal for evaluating individual columns within a training dataset.
- **D'Agostino's \$K^2\$ Test (Normaltest):** Given the large scale of our banking dataset, we also implemented this omnibus test. It works by transforming the sample skewness and kurtosis, then combining them into a single score to assess the overall shape of the distribution simultaneously.

3.Outlier analysis:

Histogram: only good for 2 or 3 features but here we have 14 to 17 features.

z-score : it is used for normal data and any observation with a Z-score absolute value greater than 3 ($|Z| > 3$) is identified and classified as an outlier.

Mahalanobis Distance : The mathematical formula for Mahalanobis Distance is based on the mean and the covariance matrix, so it is very sensitive on normality of data.

We use IQR method for recognition and mode for filling the outliers :

Robustness to Non-Normal Distributions (Non-parametric): Unlike the Z-Score method, the Interquartile Range (IQR) technique makes no assumptions regarding the normality of the data, as it operates entirely on the median and quartiles. Given that your features exhibit significant skewness, IQR serves as the optimal univariate approach.

Smart Imputation with the Median: Instead of dropping rows containing outliers—which would result in an unnecessary loss of other critical missing values—the outlying points were imputed using the median of their respective columns. Unlike the mean, the median remains resilient against the influence of extreme values, rendering it the safest statistical estimate for imputation.

We use median for outlier numeric columns and used mode for categorical columns.

4.Feature encoding:

1. Ordinal Encoding for Educational Level (0 to 5 Digits)Dimension Reduction:

By mapping these categories to ordered integers from 0 to 5, we preserve this natural ranking. This allows the machine learning models (especially tree-based algorithms like Random Forest) to understand that a 'Doctorate' (5) represents a higher level of education than 'High School' (1), capturing the underlying monotonic relationship with the credit limit.

2. One-Hot Encoding for Gender, Marital Status, and Card Category:

If we were to use simple integer encoding (like 0, 1, 2) for these columns, a regression model would mistakenly assume a numerical order (e.g., treating Category 2 as twice the value of Category 1). One-Hot Encoding solves this by creating separate binary columns (0 or 1) for each unique category. This allows the model to evaluate the presence or absence of each characteristic independently, preventing the introduction of false mathematical hierarchies.

3. Midpoint Imputation for Income Category

- If we treat these brackets as simple text or ordinal numbers (0, 1, 2...), we completely lose the actual scale and continuous nature of economic wealth. By converting each range into its mathematical **midpoint** (e.g., mapping 40K\$ - 60K\$ to 50,000\$), we transform a rigid categorical string back into a continuous, meaningful numerical feature.
- This provides the regression model with real monetary values to calculate slopes and splits, drastically improving its ability to capture the direct correlation between a customer's purchasing power and their ultimate credit limit.

There is no need for dimension reduction because the features are not that many and they are kind of related to each other.

5.Feature scaling:

Data is not normal so we use minmaxscaler

6. Model Selection Process

In this project, predicting the `Credit_Limit` of bank customers is a **Regression** task. To find the most accurate and robust model, we adopt an empirical approach by implementing and comparing three distinct types of algorithms:

1. **Simple Linear Model:** Linear Regression
2. **Regularized Linear Model:** Ridge Regression
3. **Non-Linear Ensemble Model:** Random Forest Regressor

Our selection strategy involves training these models on the preprocessed training dataset (`x_train_model`, `y_train_final`) and evaluating their performance on the unseen test dataset (`x_test_model`, `y_test_final`). The model that demonstrates the highest R-squared (R^2) score and the lowest error metrics (MAE and RMSE) will be selected as our final production model.

A. Linear Regression

Linear Regression assumes that a linear relationship exists between the input features (X) and the target variable (y). The model attempts to find the optimal hyperplane (line of best fit) by minimizing the sum of squared vertical distances (residuals) between the observed data points and the predicted line.

The mathematical equation is represented as:

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$$

- **Strengths:** Extremely fast to train, computationally efficient, and highly interpretable. The coefficients (β) directly indicate the feature importance and direction of impact on the credit limit.
- **Weaknesses:** It struggles heavily if the data contains complex, non-linear relationships (e.g., the non-linear interaction between age, card category, and income), leading to **Underfitting**.

B. Ridge Regression

Ridge Regression is an advanced variant of linear regression that incorporates **L2 Regularization**. When dealing with multiple features that might be correlated (multicollinearity), a standard linear regression can assign massive, unstable weights to features, leading to overfitting. Ridge resolves this by adding a penalty proportional to the square of the magnitude of the coefficients to the loss function:

$$Loss = \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^n (\beta_j)^2$$

- **Strengths:** Effectively prevents **Overfitting**, smooths out the impact of correlated features, and provides much more stable predictions than standard Ordinary Least Squares (OLS) regression.

- **Weaknesses:** Since it is fundamentally a linear model, it still cannot capture complex non-linear patterns inherently present in customer behavioral data.

C. Random Forest Regressor

Random Forest is a powerful **Ensemble Learning** algorithm based on the concept of **Bagging** (Bootstrap Aggregating). Instead of relying on a single Decision Tree, it constructs hundreds of independent decision trees during training. Each tree is trained on a random bootstrap sample of the data and a random subset of features. The final prediction of the Random Forest is the mathematical average of the predictions made by all individual trees.

- **Strengths:** Exceptionally capable of capturing highly non-linear relationships and complex interactions between features without requiring manual feature transformations. It is highly robust to outliers and does not assume a normal distribution of data.
- **Weaknesses:** It acts as a "Black Box," meaning it is less interpretable than linear models. It also requires more memory and computational power to train.

Model Evaluation Metrics:

To scientifically compare the performance of these models, we utilize three standard evaluation metrics:

1. Mean Absolute Error (MAE)

MAE measures the average absolute magnitude of the errors in a set of predictions, without considering their direction.

$$\text{MAE} = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i|$$

Interpretation: It gives us a direct, real-world understanding of how many dollars, on average, our model's credit limit prediction deviates from the actual credit limit.

2. Root Mean Squared Error (RMSE)

RMSE is the square root of the average of squared differences between prediction and actual observation.

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2}$$

Interpretation: Because the errors are squared before they are averaged, RMSE penalizes larger errors more heavily. A high RMSE relative to MAE indicates that the model made some severely inaccurate predictions on certain test samples.

3. R-squared Score (Coefficient of Determination)

The R^2 score represents the proportion of variance in the dependent variable (`Credit_Limit`) that can be predicted from the independent variables.

$$R^2 = \frac{SS_{res}}{SS_{tot}}$$

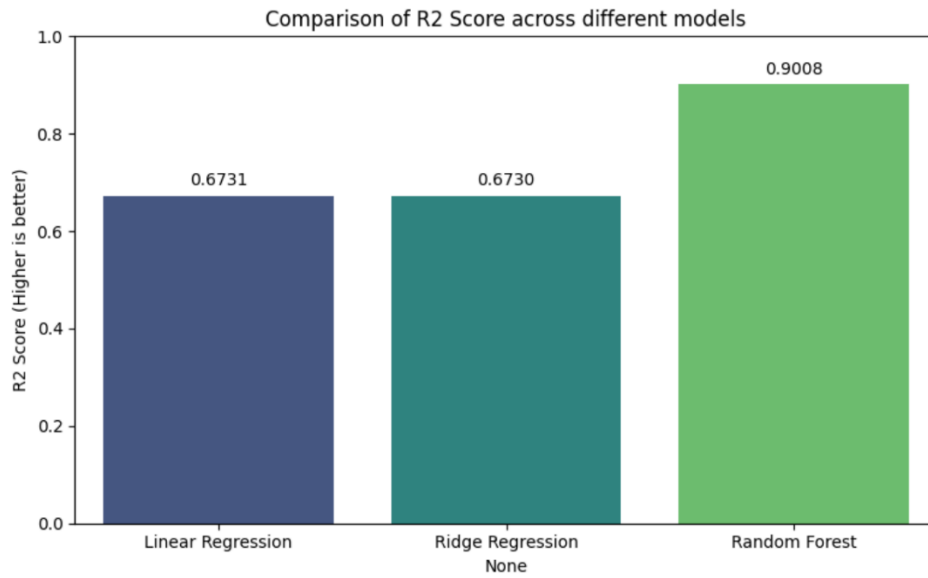
$$SS_{res} = \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$SS_{tot} = \sum_{i=1}^m (y_i - \bar{y})^2$$

Interpretation: It scales from 0 to 1 (or 0% to 100%). For instance, an R^2 score of 0.88 implies that our model successfully explains 88% of the variability in the customers' credit limits. Higher is always better.

Conclusion: Due to the presence of categorical variables (such as `Card_Category` and `Income_Category`) and non-linear dynamics within bank customer behavior and comparing results of the code, the **Random Forest Regressor** is the best fit for our data.

For dimension reduction random forest automatically chooses the features that are more important so we don't need it.



Feature importance:

